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Division:A2

Subject: Programming with Java

ASSIGNMENT No : 6

TITLE :Write a Java program to demonstrate the use of an inner class, an anonymous class.

Problem Statement

Write a Java program to demonstrate the use of an inner class, an anonymous class.

Learning Objectives

To implement inner classes and anonymous classes for improving encapsulation and event handling in Java.

Theory

An inner class is a class defined inside another class and has access to all members of the enclosing class,

including private members. Inner classes help improve code organization and logical grouping of related

classes. Anonymous classes are unnamed inner classes that are instantiated only once and are mainly used for

implementing interfaces or abstract classes. They simplify code by eliminating the need for separate class

definitions. These concepts are widely used in Java GUI programming and event handling

Outcome

Conclusion

Exercise

1.Create a Vehicle program where an inner class displays vehicle details and anonymous class performs an action.

```
class Vehicle {

    String name = "Car";
    int speed = 120;

    // Inner Class
    class Details {
        void display() {
            System.out.println("Vehicle Name : " + name);
            System.out.println("Speed      : " + speed + " km/h");
        }
    }

    void performAction() {

        // Anonymous Class
        Runnable action = new Runnable() {
            public void run() {
                System.out.println("Vehicle is moving...");
            }
        };

        action.run();
    }

    public static void main(String args[]) {

        Vehicle v = new Vehicle();
    }
}
```

```
Vehicle.Details d = v.new Details();  
d.display();  
  
v.performAction();  
}  
}
```

2. Develop a Food Delivery Application where an inner class handles order details and anonymous classes handle delivery status updates.

```
class FoodDelivery {  
  
    String customer = "Rahul";  
    String food = "Pizza";  
  
    // Inner Class  
    class Order {  
        void displayOrder() {  
            System.out.println("Customer : " + customer);  
            System.out.println("Food    : " + food);  
        }  
    }  
  
    void deliveryStatus() {  
  
        // Anonymous Class  
        Runnable status = new Runnable() {  
            public void run() {  
                System.out.println("Order is Out for Delivery.");  
            }  
        }  
    }  
}
```

```
};
```

```
status.run();
```

```
}
```

```
public static void main(String args[]) {
```

```
FoodDelivery fd = new FoodDelivery();
```

```
FoodDelivery.Order o = fd.new Order();
```

```
o.displayOrder();
```

```
fd.deliveryStatus();
```

```
}
```

```
}
```